BTS–IV(S)–04.22–0253 Reg. No. [ ][ ][ ][ ][ ][ ][ ][ ]

B

B.Tech. Degree IV Semester Supplementary Examination April 2022

ME 15-1406 MANUFACTURING PROCESSES

(*2015 Scheme*)

Time: 3 Hours Maximum Marks: 60

PART A

(Answer *ALL* questions)

$(10 \times 2 = 20)$

I. (a) Explain the process of core making in sand casting.

(b) Write short note on casting metallurgy

(c) What are allowances which are given on pattern?

(d) Explain the process of continuous casting.

(e) What are the advantages of wax casting?

(f) Differentiate between blanking and punching.

(g) What are the methods to avoid spring back in bending operation?

(h) What are the advantages of forging operation?

(i) Explain the function of electrode coating in arc welding.

(j) Which welding process is suitable for high deposition rate and speed? Justify the answer.

PART B

$(4 \times 10 = 40)$

II. (a) What do you mean by gating system in sand casting? With figure, explain all the gating elements. (5)

(b) What are the ideal characteristics of molding sand? Explain any two sand testing methods. (5)

OR

III. (a) Explain with figure, the methods to achieve directional solidification. (5)

(b) What is the function of riser in sand casting mold? Explain different types with figure. (5)

IV. (a) State any ten casting defects along with their causes and preventive measures. Use figures as necessary. (8)

(b) What is casting yield? (2)

OR

V. (a) Explain the process hot chamber and cold chamber die casting with figure. (7)

(b) Compare the advantages and disadvantages of hot chamber and cold chamber die casting. (3)

VI. (a) What do you mean by Bulk deformation process? (2)

(b) Explain the four microscopic stages of sheet metal cutting with figure. (8)

OR

VII. (a) Explain different roll arrangements used for rolling operation with figure. State the advantages of each arrangement over other. (8)

(b) Differentiate between direct and indirect extrusion process. (2)

VIII. (a) Explain the process of Laser beam welding with figure. (5)

(b) Explain any five types of welding defects. (5)

OR

IX. (a) Explain the term weldablity? (2)

(b) Differentiate between TIG and MIG welding process? (3)

(c) Which welding process is ideal in repairing railway tracks? Explain the process. (5)

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